

Curriculum Vitae

Imran Khan
Affiliated Faculty

Contact Information

South Asia Center, University of Pennsylvania
College of Arts and Sciences
255 South 36th Street, 820 Williams Hall
Philadelphia, PA 19104

Email: iuk7ad@pm.me
GitHub: github.com/iuk7ad
ORCID: orcid.org/0000-0002-1201-8107
Website: iuk7ad.github.io

Teaching Interests

Artificial intelligence, machine learning, programming, scientific models, statistics

Education

Rice University 2020-2023
Jones Graduate School of Business
Master of Business Administration (MBA)
Capstone: Play Forever (Canada)

New York University 2017-2019
Tandon School of Engineering, Center for Urban Science and Progress
Master of Science (MS) - Applied Urban Science and Informatics
Capstone: Quantifying segregation in New York City schools (with machine learning clustering)

University of Virginia 2011-2013
Batten School of Leadership and Public Policy
Master of Public Policy (MPP)
Capstone: Green procurement strategic plan: Policy analysis, planning, and implementation of sustainable New York City operations

Southern Methodist University 2009-2010
Simmons School of Education and Human Development, Department of Teaching and Learning
Master of Education (MEd) w/ Honors
Thesis: Evaluation and implications of drop-out prevention program among minority adolescents

Texas A&M University 2000-2003
College of Agriculture and Life Sciences, Department of Biochemistry and Biophysics
Bachelor of Science (BS) w/ Honors - Biochemistry
Thesis: Protein peptide purification of Human Metal-responsive Transcription Factor 1

Teaching Experience

Community College of Philadelphia 2024-2025

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Courses taught: Database Management Systems, Foundations of Data Science, Introduction to Information Technology, Machine Learning

City University of New York - Hunter College 2022-Present
Course taught: Relational Databases and SQL Programming

The Carpentries 2022-Present
Courses taught: Data Carpentry, Library Carpentry, Software Carpentry

City University of New York - Energy Management Institute 2020-Present
Course taught: Problem Solving Using Microsoft Excel

City University of New York - LaGuardia Community College 2020-Present
Courses taught: Data Analytics Foundations, Data, Databases and SQL, Data Visualization with Tableau, Power BI for Data Analytics, Predictive Analytics: Machine Learning, Python for Data Analytics

Garland Independent School District 2010-2011
Course taught: Chemistry

Collin College 2010
Course taught: Chemistry

Pegasus School of Liberal Arts and Sciences 2009
Course taught: Chemistry

Pasadena City College 2007
Course taught: Chemistry

California State University, Los Angeles 2004-2006
Course taught: Chemistry

Research Experience

Texas A&M University 2025-Present
High Performance Research Computing

New York University 2019
Urban Modeling Group

New York University 2018
GovLab

Southern Methodist University 2010-2011
John Goodwin Tower Center for Political Studies Student Forum

Curriculum Vitae

Texas A&M University 2003
United States Department of Agriculture, Agricultural Research Service

Texas A&M University 2001-2004
College of Agriculture and Life Sciences, Department of Biochemistry and Biophysics

Industry Experience

CivicReach 2026-Present
Government Partnerships Broker

Hudson Shipping Corporation 2025-2026
Co-Founder and Chief Operating Officer

LaGuardia APEX 2024
Procurement Specialist

Cyber Agility Academy 2024-2025
Co-Founder and Chief Operating Officer

Ernst and Young Services Ltd. 2022-2024
Engagement Manager (Financial Services, Extended Assurance, Internal Audit, Digital Risk and Innovation)

Atkinson-Noland and Associates 2020
Engineer
Certification: OSHA Construction (30-Hr)

Mumble Solutions 2019
Software Engineer - Lead Developer

Arcadis 2018
Data Scientist

Build Well Labs 2018-2026
Principal
Clients: Datapolitan, Education Pioneers, Egerton Clinic, Healthfirst, Movement Labs, New York Academy of Sciences, New York City Department of Education, Philosophy LLC, United States Department of Defense, United States Department of Health and Human Services
Pro Bono Clients: American Chemical Society, Code for America, Company OnStage, Cooking for Community, DataKind, Dusit International, Mandela Washington Fellowship for Young African Leaders, New Politics Leadership Academy

Acumen Solutions 2013
Consultant

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Chemical Information Services 2009-2010
Chemical Data Researcher

Teva Pharmaceuticals (formerly Watson Pharmaceuticals) 2008
Scientist
Certification: CGMP (Current Good Manufacturing Practices)

3M 2007
Materials Scientist

Curbside, Inc. 2007
Chemist and Hazardous Materials Specialist
Certification: OSHA HAZWOPER (40-Hr)

Government Experience

New York City Mayor's Office 2016-2017
Office of Operations, Office for Economic Opportunity
Director of Special Projects
Assignments: Education, Housing Authority, Service, Social Services, Young Men's Initiative,
Youth and Community Development

Federal Bureau of Investigation 2015
Citizens Academy

New York Police Department 2015
Citizens Police Academy

New York State Governor's Office 2013-2015
Department of Agriculture and Markets, Division of Agricultural Development
Empire State Fellow
Assignments: Anti-Hunger Task Force, Community, Opportunity, Reinvestment Neighborhoods
Initiative, Lean Spending and Government Efficiency Commission, Long Island, Mid-Hudson,
and New York City Regional Economic Development Councils, Office of General Services,
Rochester Anti-Poverty Task Force

Fellowships

United States Army 2022
DEVCOM Army Research Laboratory
Foundry Program via FedTech, National Security Innovation Network

Rice University 2021-2022
Fondren Library
Fondren Fellowship
Certification: Software Carpentry Instructor

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United States Department of Commerce Census Bureau Civic Digital Fellowship via Coding it Forward	2020
University of Southern California Dornsife College of Letters, Arts and Sciences, Center for Religion and Civic Culture American Muslim Civic Leadership Institute Fellowship	2018
New Leaders Council Capital District New York Fellowship	2014
State University of New York Rockefeller Institute of Government Empire State Fellowship	2013-2014
New York City Department of Citywide Administrative Services Office of Citywide Procurement Green Procurement Fellowship	2013
United States Department of Energy Office of Economic Impact and Diversity Minority Educational Institution Student Partnership Program Fellowship	2012
Houston Independent School District School Improvement Office Education Pioneers Summer Graduate School Fellowship	2011
California Institute of Technology Division of Chemistry and Chemical Engineering National Science Foundation Partnership for Research and Education in Materials Fellowship	2004-2007
<u>Service Projects</u>	
Delaware Division of Libraries Dover, DE	2024-2025
BakerRipley Houston, TX	2020-2021
New York City Mayor's Office of Criminal Justice New York, NY	2018
New York City Department of Education New York, NY	2015

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CitySquare Dallas, TX	2010
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Board Appointments

LaGuardia Community College Artificial Intelligence Certification Program AI Advisory Board Member	2026
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Service Year Alliance Alums Amplified Speakers Bureau Member	2025-Present
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University of Pennsylvania, College of Arts and Sciences South Asia Center Affiliated Faculty Member	2025-Present
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LaGuardia Community College Generative AI Working Group Member	2025-2026
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LaGuardia Community College AI Advisory Board Member	2024-2025
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Harvard Business Review Advisory Council Member	2024-Present
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American Bar Association Model Procurement Code Revision Drafting Committee Member	2024-Present
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techUK TechSkills Data Science Working Group Member	2023-2024
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Cyber Readiness Institute Small Business Advisory Council Member	2023-Present
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American Chemical Society, New York Local Section Computers in Chemistry Topical Group Co-Chair	2019-2020
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New York City Human Resources Administration SNAP Task Force Member	2015
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Bedford Stuyvesant Restoration Corporation Central Brooklyn Food Hub Task Force Member	2015
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Cornell Small Farms Program 2015
Northeast Sustainable Agriculture Research and Education Advisory Board Member

American Farmland Trust 2014-2015
Farm to Institution New York State Leadership Team Member

New York State Energy Research and Development Authority 2014
Technical Evaluation Panel Member

Current Professional Association Membership

American Chemical Society

Association for Supply Chain Management

Engineers' Club of Philadelphia

Research Data Alliance

United States Digital Response

United States Research Software Engineer Association

Publications and Reports

Bateman, A., Grove, L., Swartz, H., and Yang, Z., and Khan, I. "Linking Long Island agriculture to New York City public schools," University of Virginia Frank Batten School of Leadership and Public Policy, 2015.

Dudley, R., Goodman, D., Hunt, J., and Jung, J., and Khan, I. "Growing local food systems: Policies to expand market access for New York's small farmers," University of Virginia Frank Batten School of Leadership and Public Policy, 2014.

Connors, K., Khan, I., Orphanides, E., and West, M. "Income-generating programs: Best practices and recommendations for Lebanon," University of Virginia Frank Batten School of Leadership and Public Policy, towards partial fulfilment of Master of Public Policy coursework and degree requirements, 2012.

Mathias, E.V., Liu, X., Franco, O., Khan, I., Ba, Y., and Kornfield, J.A. "A model of drug-loaded fluorocarbon-based micelles studied by electron-spin induced ^{19}F relaxation NMR and molecular dynamics modeling," *Langmuir*, 24, 692-700, 2008.

Conference Invitations and Talks

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Texas A&M University, Division of Research, High Performance Research Computing, Building Research Innovation at Community Colleges Research Data Management Conference 2026, March 2026, Alexandria, VA

University of Pennsylvania, Data Carpentry, March 2026, Philadelphia, PA

University of Massachusetts Dartmouth School for Marine Science and Technology, Software Carpentry, October 2025, Dartmouth, MA

Texas A&M University, Division of Research, High Performance Research Computing, Summer PACES Research Training Workshop, August 2025, Albuquerque, NM

Texas A&M University, Division of Research, High Performance Research Computing, Building Research Innovation at Community Colleges Research Data Management Conference 2025, July 2025, Alexandria, VA

University of Massachusetts Dartmouth School for Marine Science and Technology, Data Carpentry, November 2024, Dartmouth, MA

University of Hawaii, Data Carpentry, October 2024, Honolulu, HI

University of Pennsylvania, Data Carpentry, June 2024, Philadelphia, PA

Pennsylvania State University, Software Carpentry, April 2024, State College, PA

Michigan Technological University, Data Carpentry, February 2024, Houghton, MI

University of Sussex, Data Carpentry, July 2023, Brighton, England

University of Pennsylvania, Penn Graduate Consulting Club, November 2022, Online

University of Pennsylvania - Van Pelt Library, Data Carpentry, October 2022, Philadelphia, PA

The i4 Group, Data-Driven Service Development, June 2022, Online

The i4 Group, Using Open Data for Public Benefit, March 2022, Online

The i4 Group, Data Driven Product Development, August 2021, Online

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General Assembly, Data Week | Let's Break it Down: Data Science vs. Data Analytics, May 2021, Online

Long Island Food Conference, April 2015, Hempstead, NY

Farm to Institution Summit, April 2015, Amherst, MA

AcademyHealth, National Health Policy Conference, February 2015, Washington, DC

Cornell University, Cornell Cooperative Extension, New York Farms to New York Plates: The Becker Forum on Institutional Purchasing Across NYS, January 2015, Syracuse, NY

American Farmland Trust, New York State Office, Hudson Valley Farm to Institution Summit, October 2014, New Paltz, NY

Heritage Radio Network, The Future of Food Leadership: Feeding NYS, September 2014, Brooklyn, NY

Empire Farm Days, August 2014, Seneca Falls, NY

White House, Summit on Working Families, June 2014, Washington, DC

New York State Office of General Services, Purchasing Forum and Trade Show, May 2014, Albany, NY

SOMOS El Futuro, Spring Conference, March 2014, Albany, NY

New Leaders Council, Northeast Conversation, March 2014, Jersey City, NJ

Asian American Government Executives Network, Leadership Workshop, July 2013, Arlington, VA

StartingBloc, Institute for Social Innovation, February 2013, Los Angeles, CA

International Sociological Association RC28 Conference, Labor Market and Educational Transitions in Uncertain Times, August 2012, Charlottesville, VA

Conference on Asian Pacific American Leadership, Washington Leadership Program, Summer 2012, Washington, DC

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Clinton Global Initiative, Clinton Global Initiative University, March 2012, Washington, DC

National Academy of Social Insurance, Social Insurance in a Market Economy: Obstacles and Opportunities, January 2012, Washington, DC

Khan, I. "Develop U: Empowering students, families, and communities through education," October, 2010.

Khan, I. and Springer, K. "Evaluation and implications of drop-out prevention program among minority adolescents," December, 2009.

Khan, I. "The SIOP model: Making content comprehensible for English language learners," July, 2009.

Khan, I. "How MAS Youth fulfills the needs of young people," October, 2008.

Khan, I. "MAS Youth worker orientation and training," July, 2008.

Khan, I. and Ba, Y. "Computational molecular dynamics studies of drug-loaded fluorocarbon-based micelles," May, 2007.

Khan, I. and Ba, Y. "Structure and function of zinc finger proteins," May, 2006.

Khan, I. and McCurdy, A. "Design of a novel transition metal-binding spiropyran catalyst for use in asymmetric synthesis," February, 2005.

Khan, I. and Giedroc, D. "Protein peptide purification of Human Metal-responsive Transcription Factor 1," August, 2003.

Awards and Honors

Rice University Jones School Dean's Scholarship Award, 2020

Islamic Scholarship Fund Ilhan Omar Courage and Leadership Scholarship, 2018

Islamic Scholarship Fund Deah Yusor Razan Scholarship, 2017

New York City Mayor's Graduate Scholarship, 2017

Kresge Foundation Population Health Scholarship, 2015

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New York State Canal Ambassador, 2014

Asian American Government Executives Network Scholarship, 2013

University of Virginia Batten Professional Development Grant, 2011

University of Virginia Batten Merit-Based Fellowship, 2011

Segal AmeriCorps Education Award, 2010

United States Department of State International Visitor Leadership Program Host, 2008

Robert A. Welch Foundation Scholarship, 2003

Texas A&M University Agriculture and Life Sciences Dean's Honor Role, 2001

Texas A&M University Agriculture and Life Sciences Distinguished Student, 2000

Skills and Techniques

Python packages: GeoPandas, Matplotlib, NumPy, Pandas, Plotly, Scikit-learn, Seaborn, StatsModels

Coding and programming languages/software: Amazon AWS Redshift, ArcGIS, AutoCAD, C++, CARTO, Cerius2, CHARMM, Empower, Gaussian, Insight II, LaTeX, MATLAB, Microsoft Power BI, Microsoft SQL Server, PostgreSQL, R, Salesforce, Stata, Tableau, UiPath Process Mining

Microsoft Office Excel plug-ins: Oracle Crystal Ball, Rutgers YASAI, and Solver

Instrumentation and lab techniques: Column chromatography, Dissolution, Electron microscopy (SEM/TEM), EPR/NMR spectroscopy, HPLC/GC/MS, LIDAR, Materials failure analysis, Metallurgy, Organic synthesis, PCR, Polymer design, and Protein purification